

UMD professor shows how to turn PlayStation 3 into supercomputer

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DARTMOUTH — First he stunned the science world by building a supercomputer for only a few thousand dollars, using parts from the Sony PlayStation 3.

Now he's sharing the secret.

In a new online manual, UMass Dartmouth physics professor Gaurav Khanna explains step by step how to retool the popular video game system into a sophisticated computer capable of solving complex equations in seconds.

As to why he spilled the beans, Dr. Khanna said he really had no choice. Ever since he and technician Glenn Volker rewired eight PlayStations in 2007 in an attempt to unlock the mysteries of the universe using math, he's been fielding phone calls from scientists and avid gamers who long to build supercomputers of their own.

Dr. Khanna's experiment landed him national media coverage one year ago. He said he still gets a lot of e-mails from the general public — people who have PlayStation 3's but perhaps have grown tired of playing Grand Theft Auto IV and Rock Band.

"I hear from people who say it's nice to see that something they spend time entertaining themselves with you can do real science with," Dr. Khanna said.

The manual is divided into three thorough parts, complete with screen shots of the codes and prompts people are likely to see during assembly. The Web site, which also includes a list of additional resources and frequently asked questions, was designed to make the process digestible to both scientists and amateur computer lovers.

That includes people who want to get lost in the infinity of pi and those who are into more complex problem-solving, such as predicting the properties of gravitational waves generated by black holes at the center of galaxies, which is the basis of Dr. Khanna's research.

"Everyone knows what a PlayStation is and everyone knows what a black hole is, but you don't normally put them together," Dr. Khanna said.

The online instructions, available at www.ps3cluster.org, offer advice for turning a single PlayStation 3 into a super-fast work station or using a cluster of the consoles to create a computer that can handle complex scientific research.

When Dr. Khanna started the PlayStation project in 2007, he said he was one of only two researchers in the United States to have used gaming systems as supercomputers. The other was in North Carolina,

he said.

Dr. Khanna said retooling PlayStation 3s and similar gaming consoles is the most cost-effective way to do supercomputing in the 21st century. Typically, scientists rent supercomputer time by the hour. A single simulation can cost \$5,000, not good considering science budgets have been dropping significantly the past 10 years, Dr. Khanna said. For the same cost, a homemade supercomputer made from a PlayStation 3 will work just as well, if not better, and will last much longer.

"Especially in science, simulation and computation are becoming more and more popular," Dr. Khanna said. "It's much cheaper to simulate something than it is to actually build it. More people want to look at solutions and run those simulations cheaply, and that motivated me to go the route of creating a guide."

After word spread of Dr. Khanna's discovery, the professor teamed up with Boston-based scientist Chris Poulin to conduct more advanced research. Mr. Poulin, who runs a data analysis company, had been looking for cost-effective ways to crunch numbers for his own projects at the time.

The men combined resources, cleaned up and simplified their research and realized they had enough to make a clear, comprehensive guide. They viewed it as a way to share their research while standardizing the procedures, Mr. Poulin said, adding that the manual has a twofold purpose.

"It reduces the cost of scientific computing and pushes forward computer education, instructions for hobbyists, that sort of thing," Mr. Poulin said. "It's already been a success to me because it's reduced the cost of my projects, but now that I'm sharing that information, I hope it finds an audience."

Mr. Poulin said he would like to see the technology eventually put to use in high school science classrooms.

Sharing the information opens the door to partnerships with industry and other universities, Dr. Khanna said, noting that in addition to Sony, Tyco Electronics, Terra Soft Solutions and IBM are among the companies already involved with the project. He said the UMass Dartmouth College of Engineering has an interest and focus in simulation sciences.

Dr. Khanna said Sony gave him eight PlayStations at his request in the early stages of his research last year. He said he needed at least that many to run the programs he wanted to use. Since then, the number of consoles in his lab has doubled to 16, but that number likely will not rise.

With great power comes great heat.

"It's a good number, 16. I'm going to stick with that because in the lab I have right now, they generate quite a bit of heat. Unless I get another room in the university, we'll stay with what we have."

More than a year after his research began, Dr. Khanna said people still are surprised to learn that gaming consoles are capable of performing tough equations. He said those people fail to realize that behind the flashy graphics, electronics ranging from gaming systems to high-definition cameras all rely on complex number crunching to function.

Sony has plans to market a "work station" to scientists next year with a processor and graphic

capabilities similar to that of a PlayStation, but such a unit likely will be much more expensive, Dr. Khanna said.

While the prices on some computing software are coming down, the best method remains using gaming consoles, Mr. Poulin said.

Mr. Poulin and Dr. Khanna said they are looking at the potential other gaming systems have for supercomputing and hope to add to the manual as they conduct research.

"We're hoping that as time goes on, we can add more sections. When the PlayStation 4 comes out, that's the obvious one to do," Dr. Khanna said.